



PES UNIVERSITY

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**LOW POWER VLSI
MINI PROJECT REPORT
ON**

**Leakage Minimization of 10T Full Adder Using Deep Sub
Micron Technique**

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under the guidance of

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PROGRAM B.TECH

Introduction:

The research addresses the critical challenge of leakage power dissipation in deep sub-micron (DSM) regimes, specifically focusing on 45nm CMOS technology. As digital devices scale, the reduction in threshold voltage and gate oxide thickness leads to a significant rise in sub-threshold leakage and oxide tunnelling current, which often dominate the power profile of portable electronics. Central to this study is the 1-bit 10-transistor (10T) full adder, a design chosen for its potential to optimize the speed-area-power trade-off. Unlike conventional CMOS adders that suffer from high short-circuit power, the 10T architecture minimizes direct paths between the supply and ground. The paper aims to provide a systematic evaluation of this topology by analysing its performance within an adder chain, using a novel figure of merit to account for realistic interconnection parasitic and timing dependencies that are often overlooked in theoretical models.

Methodology:

The experimental framework utilizes the Cadence Virtuoso environment with a 45nm standard CMOS process at a nominal supply voltage of 2V. The 10T adder core is implemented using five CMOS inverters, where input signals A and B are strategically routed to drive specific inverter stages, and the power supply is uniquely restricted to the first inverter to curb power dissipation. To minimize leakage, the methodology integrates process-level optimizations, such as increasing substrate doping and refining channel profiles, with circuit-level strategies including self-reverse biasing and Multi-Threshold CMOS (MTCMOS) techniques. A key technical contribution is a transistor resizing approach used to identify the optimal sleep transistor size, balancing the reduction of leakage current against area constraints. Performance metrics—including power, delay, and the power-delay product (PDP)—were rigorously simulated across 180nm, 90nm, and 45nm nodes to validate the design's scalability and efficiency in high-frequency applications.

a) Base paper circuit:

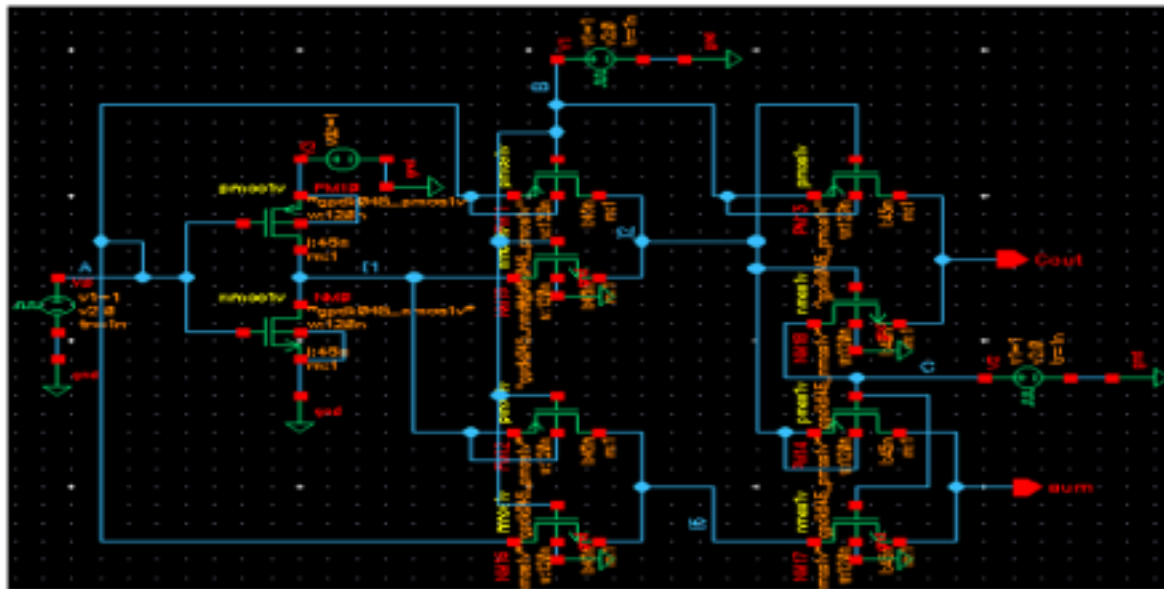
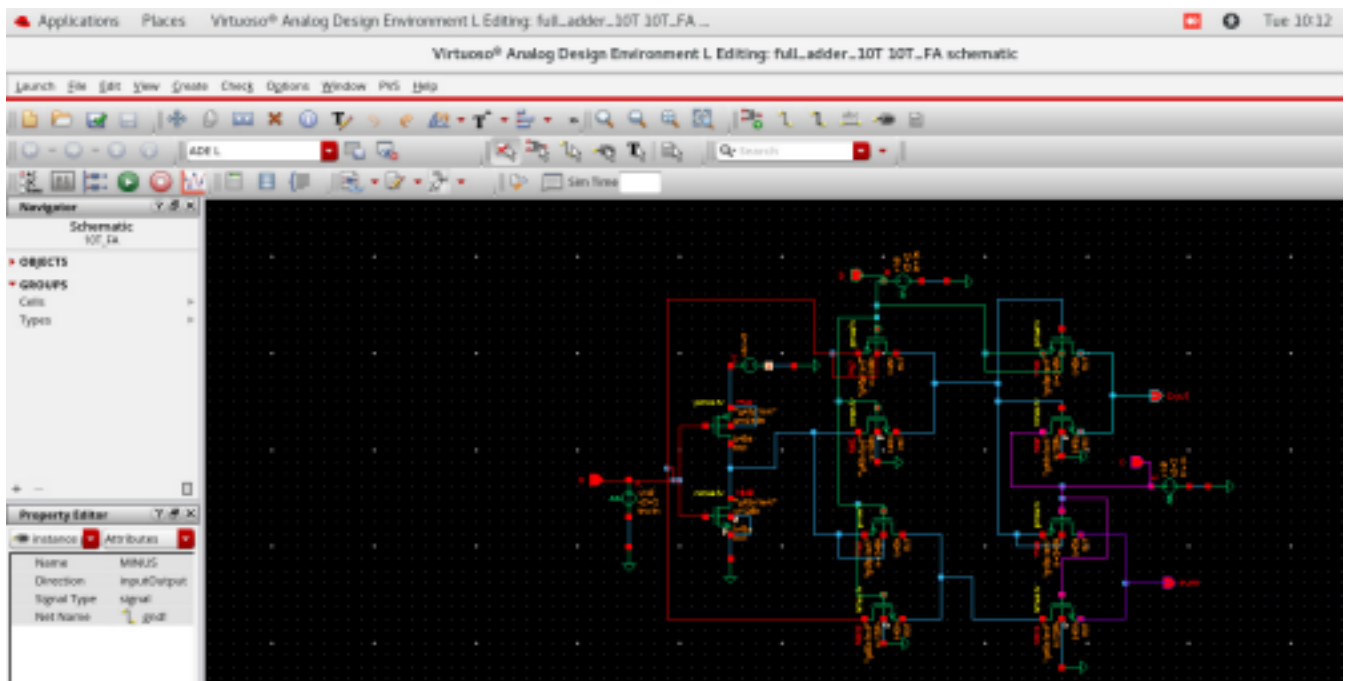


Fig-1 Ten Transistor Adder cell

b) Circuit implemented using base paper methods

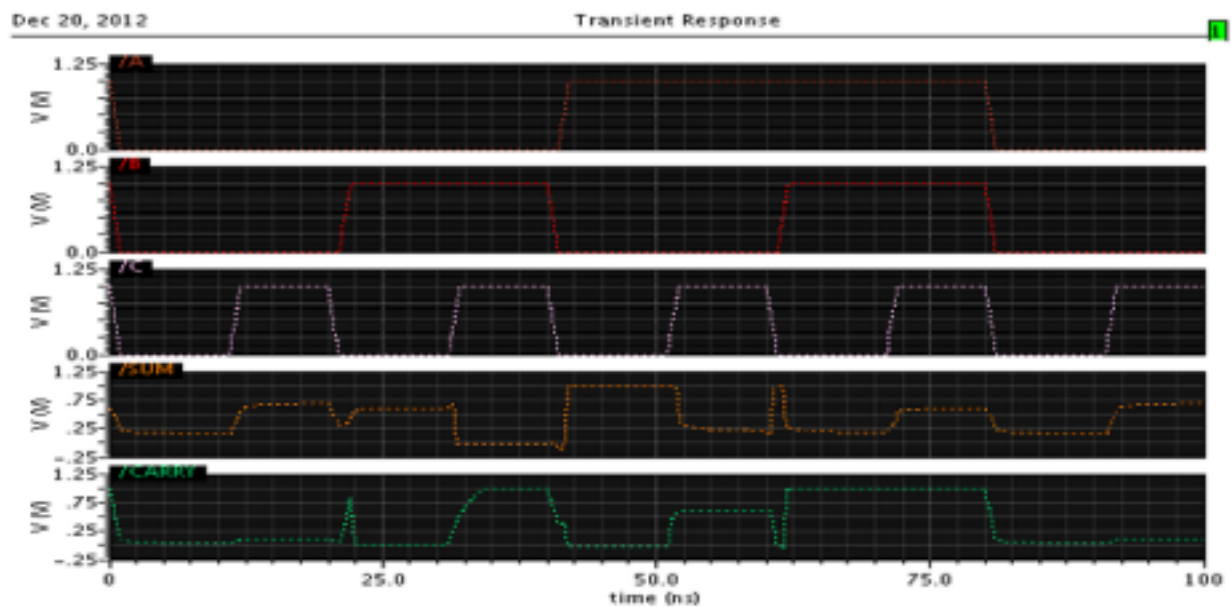


Results:

a) Base paper results

Not specified.

Base paper waveform



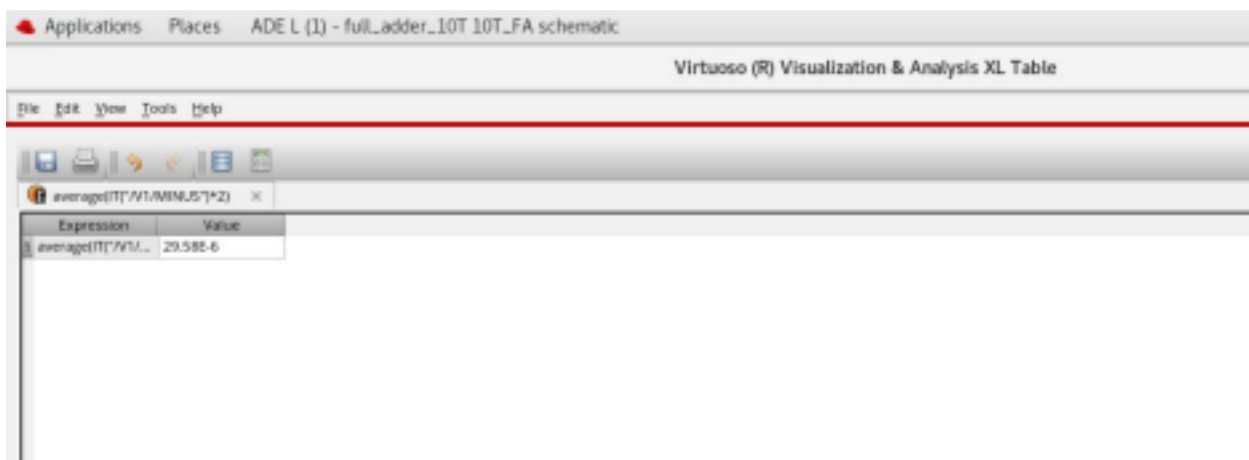
Waveform of the 10T Full adder:



a) Power achieved using the base paper circuits method

Avg ($IT("V1/MINUS")$) = 14.79E-6 A

VDD = 2V



b) Power achieved using power reducing techniques

CASE 1: Reduced VDD from 2V to 1V

Avg ($IT("V9/MINUS")$) = 4.5945E-6 A

VDD = 1V

Virtuoso (R) Visualization & Analysis XL Table

File Edit View Tools Help

average(IT*(V1/MINUS)*P2) x

Expression	Value
average(IT*(V1/MINUS)*P2)	9.189E-6

CASE 2: Replacing the tt (Typical-Typical) by sf (Slow-Fast) in ADE L model libraries.

Avg (IT("V9/MINUS") = 13.07 E-6 A

VDD = 2V

Virtuoso (R) Visualization & Analysis XL Table

File Edit View Tools Help

average(IT*(V1/MINUS)*P2) x

Expression	Value
average(IT*(V1/MINUS)*P2)	26.14E-6

CASE 3: Changing length (L) of all the mosfets used in the project from 45n M to 90n

M Avg (IT("V9/MINUS") = E-6 A

VDD = 2V

Virtuoso (R) Visualization & Analysis XL calculator

Virtuoso (R) Visualization & Analysis XL Table

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average(IT*(V1/MINUS)*P2) x

Expression	Value
average(IT*(V1/MINUS)*P2)	24.06E-6

CASE 4: Implementing all these above 3 cases at once

Avg (IT("/V9/MINUS") = E-6 A

VDD = 1V



Expression	Value
average(IT["/V1/MINUS"]*2)	7.099E-6

Conclusion:

The optimized 10T full adder reduces power from 29.58 μW to 7.099 μW , achieving nearly 76% improvement over the base design. This clearly shows that combining voltage scaling, process variation analysis, and device optimization is highly effective for low-power VLSI design.